

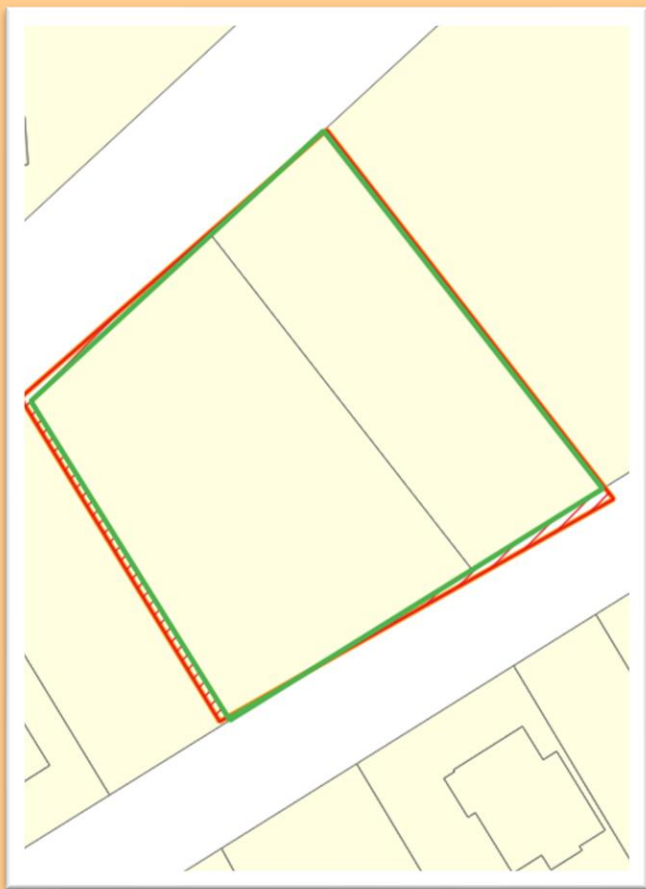
'brdr' : Alignment of Geospatial Boundaries

A Python library aligning thematic geometries to reference borders

FOSS4G.be - 25/09/2025

Karel Dieussaert

Agenda



'brdr' : Alignment of Geospatial Boundaries

- Why? The origin of the project
- What? brdr (& brdrQ)
- How? The algorithm
- Use cases

Why?

Over het onroerend goed

CaPaKey: 44045H0850/02E000

Adres: Lindenplaats 1, 9180 Moerbeke



Onroerend erfgoed

Onroerend erfgoed: 'Moervaartdepressie' - 10383

Geldig van 18/12/2009

Type: Landschapsatlasrelict, vastgesteld

<https://id.erfgoed.net/aanduidingsobjecten/10383>

bron: Onroerend Erfgoed (bevestigd 15/01/2024)

Onroerend erfgoed: 'Burgerhuizen en brouwerij' - 141655

Geldig van 20/06/2023

Type: Bouwkundig erfgoed, vastgesteld

<https://id.erfgoed.net/aanduidingsobjecten/141655>

bron: Onroerend Erfgoed (bevestigd 15/01/2024)



Data Utility Company



Exchanging **spatial** information

Where is protected heritage?

DATA SOURCES

Source X



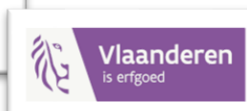
Source:
Flemish Heritage Agency

Source Z

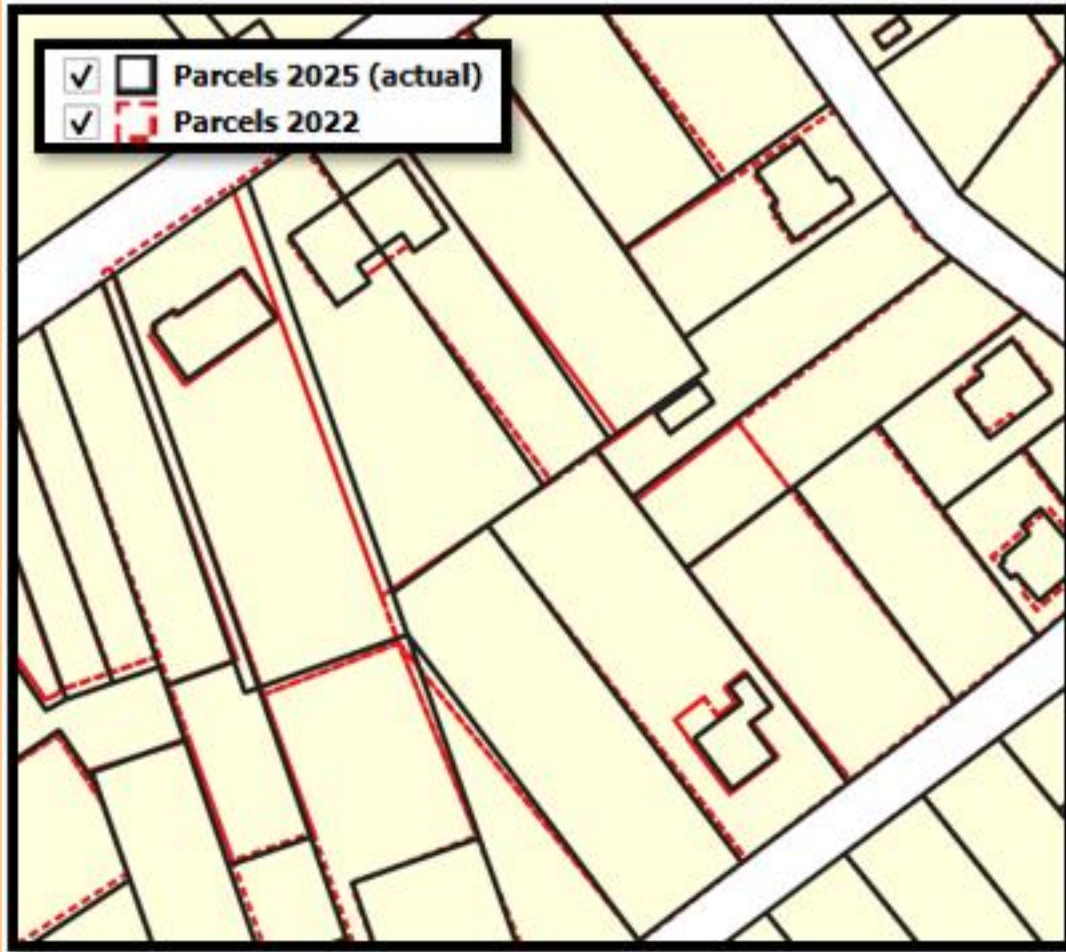
- Exchange of **spatial** information
- Queries based on **Geometry**
- Geodata **quality** is crucial (!)



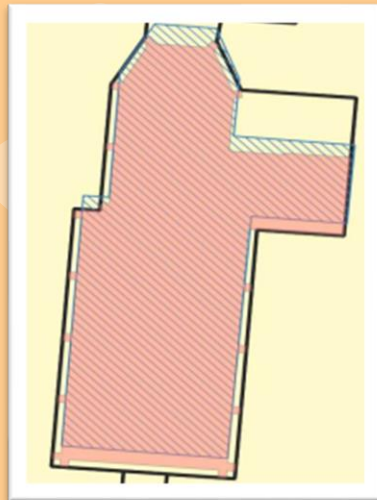
Data quality enhancement project
(partnership Athumi – Flemish Heritage Agency)



Geodata quality



Flanders Heritage geodata



↷ Very diverse

↷ Basic (simple) case:

- Contour based on 1 or more parcel

↷ Other cases:

- Contour based on other references (buildings, artwork,...)
- Contour with donuts ('inner holes')
- Contour covering 'part of parcel'
- Narrow structures (f.e hedges, facades,...)
- Contour on Public Domain (no-reference)

↷ ➔ Generic solution for Geometry Alignment: 'brdr'

What is brdr?

BRDR - a Python library to assist in realigning (multi-)polygons (OGC Simple Features) to reference borders



Python package (available on PyPi & Github)

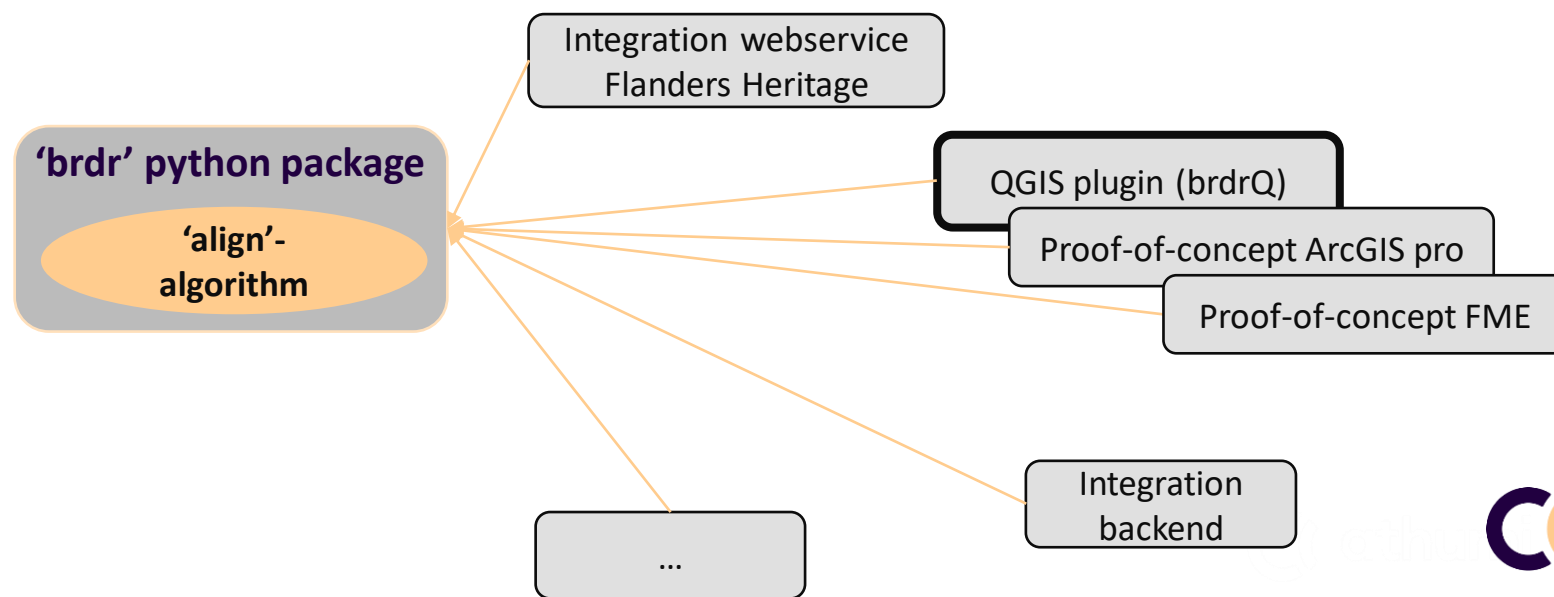
Open-source (MIT-license)

Goal:

- Re-aligning (multi-)polygons to reference borders
- Ensures consistency and quality in boundary datasets

Generic:

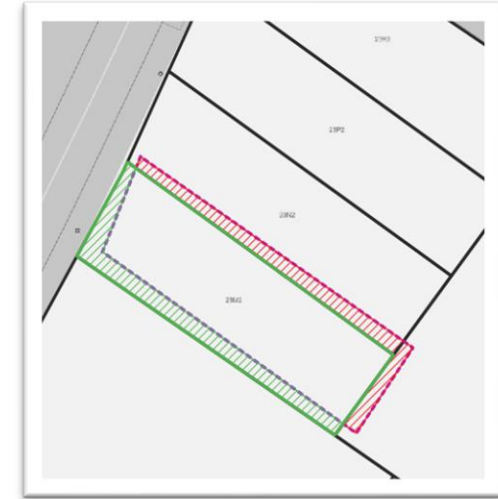
- Not tied to a region or business
- Algorithm (calculation) separated from GUI/applications



'brdr' principles



INPUT DATA(SET)



ALIGNED DATA(SET): PREDICTION(S) & DIFFERENCES

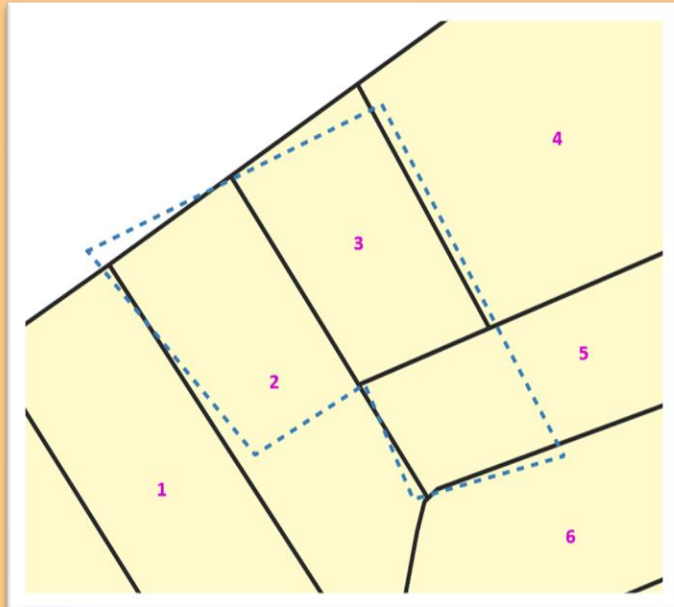
Intentionality:

- intentional deviations can be retained

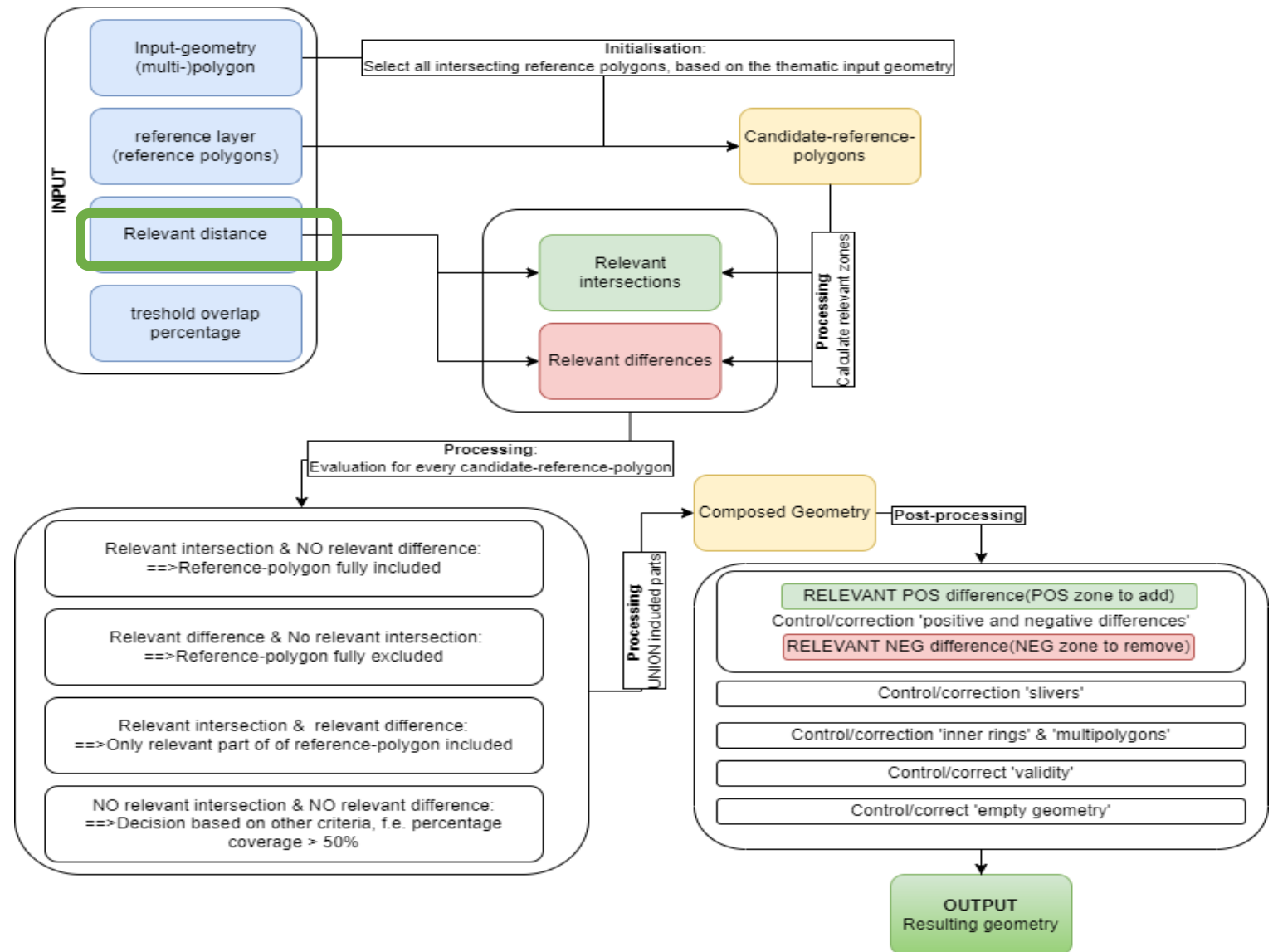
Conservation of shape:

- re-use geometry of reference borders

Alignment-algorithm ('brdr')



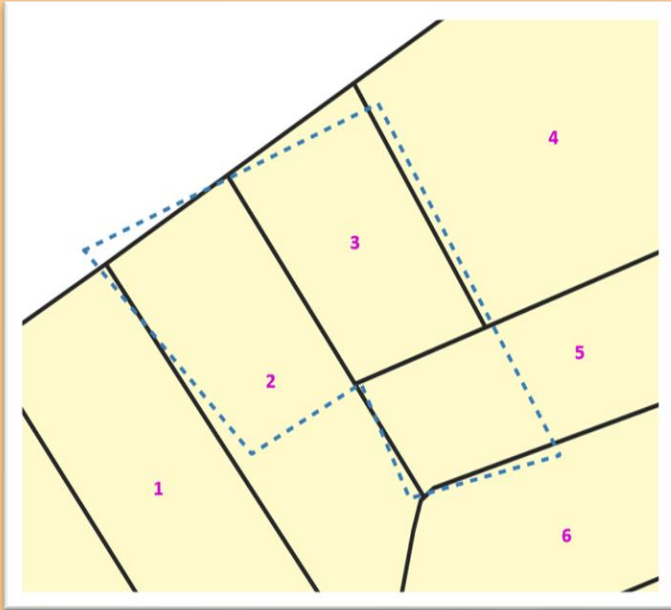
RELEVANCE =
...important, relevant (in a specific context). In short, relevant refers to a substantive contribution: a contribution that "matters"...



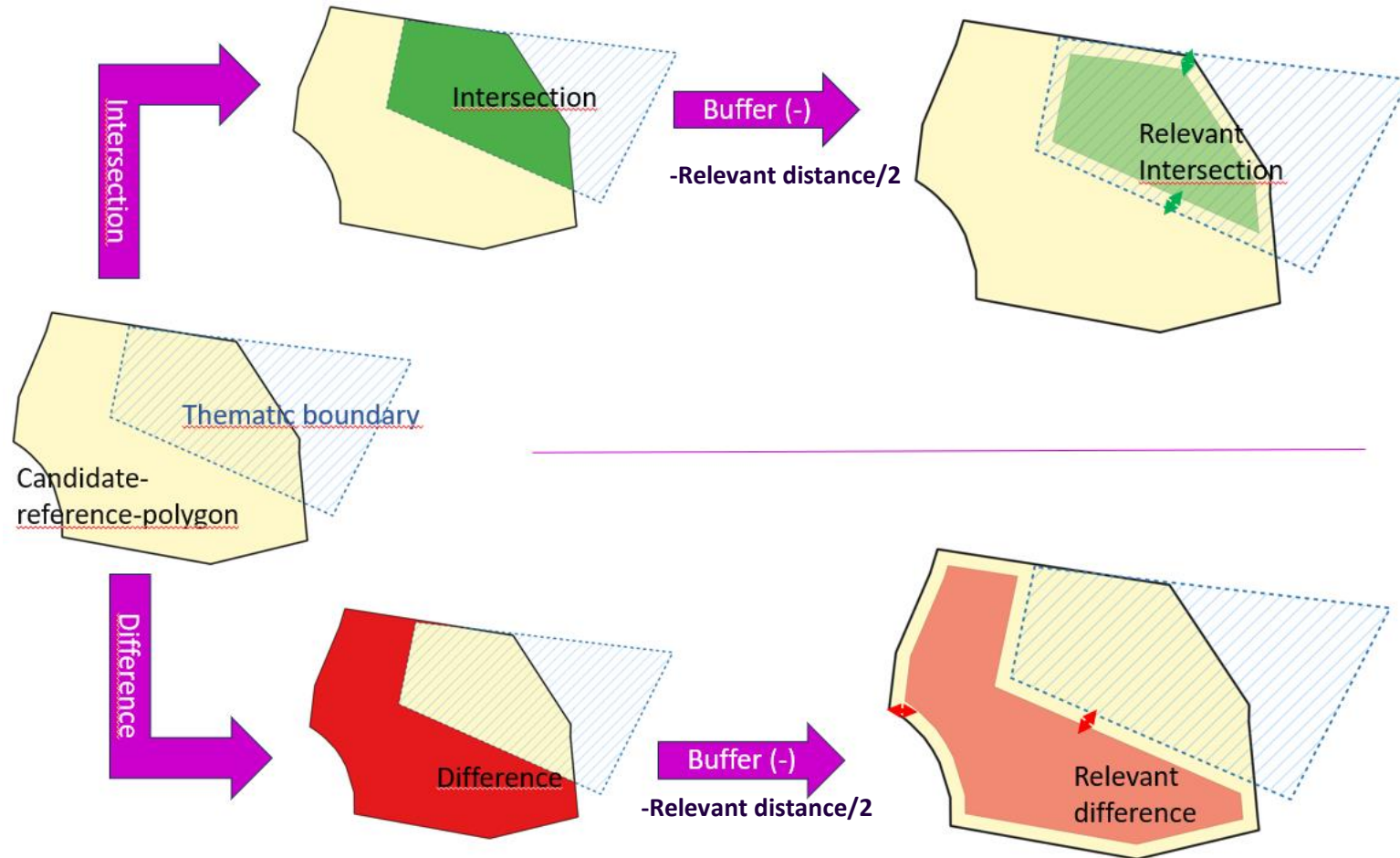
“Relevant distance”:

- Distance used to determine which zones are included or excluded in the resulting geometry.
- ~Maximum distance that the resulting geometry may change.

Alignment-algorithm (‘brdr’)

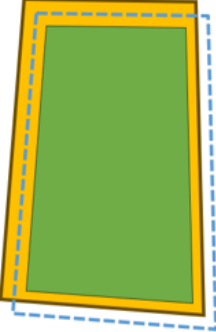
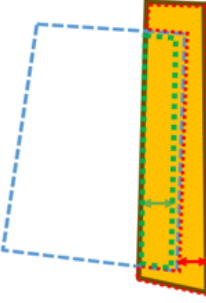
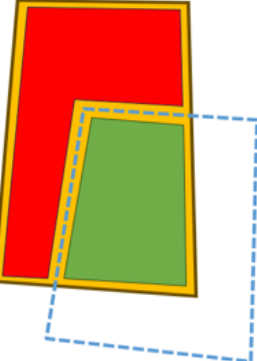
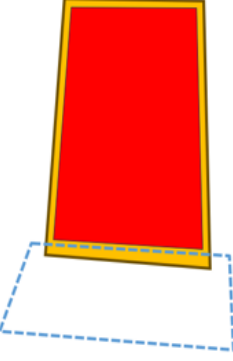


“Relevant zones”: Relevant intersection & relevant differences

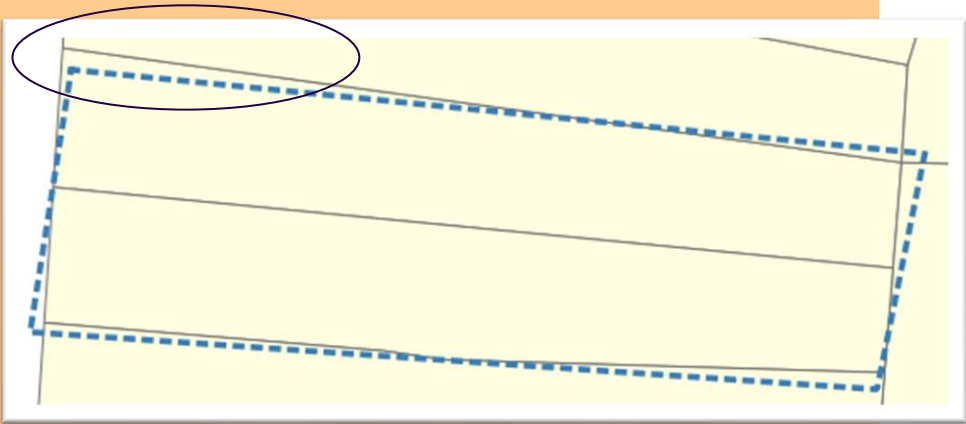


Alignment-algorithm (‘brdr’)

“Relevant intersections & relevant differences”

		Intersection	
		Relevant	Not-relevant
Difference	Not-relevant	 • Relevant intersection • No relevant difference → Parcel fully included	<i>(Bij grote relevante afstand of kleine/smalle percelen)</i>  • No relevant intersection • No relevant difference → Include or exclude parcel? → Choice based on overlap-parameter, f.e. percentage coverage > 50%
	Relevant	 • Relevant intersection • Relevant difference → Parcel partially included	 • No relevant intersection • Relevant difference → Parcel excluded

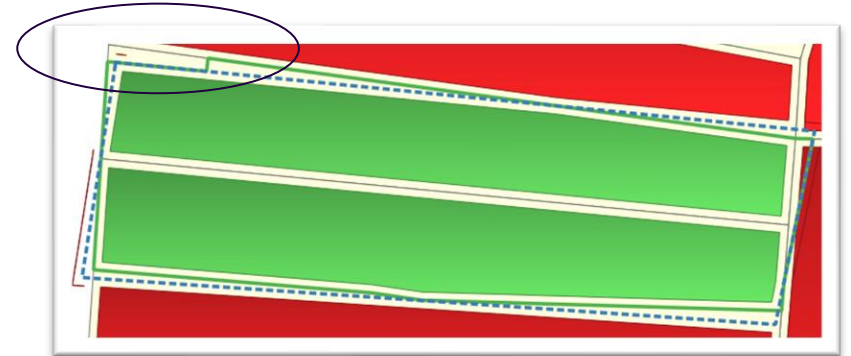
Alignment-algorithm (‘brdr’)



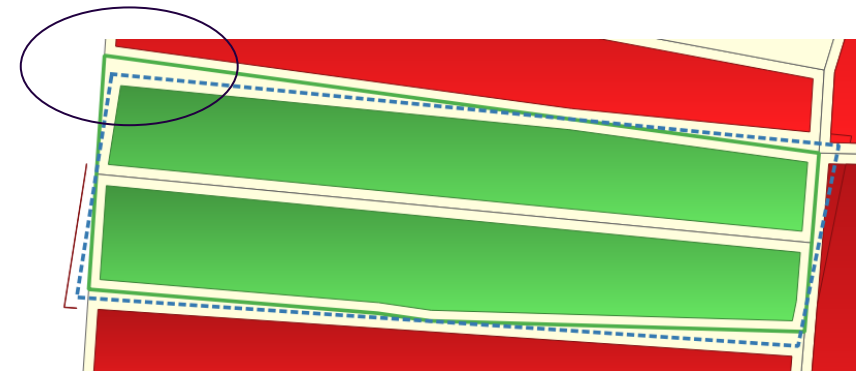
Relevant distance
0,6 m



Relevant distance
0,8 m



Relevant distance
(>) 1,0 m



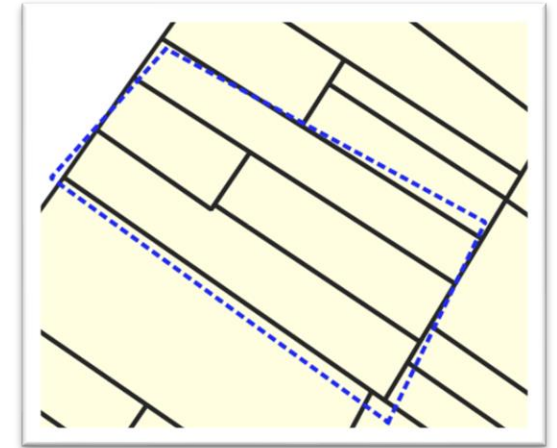
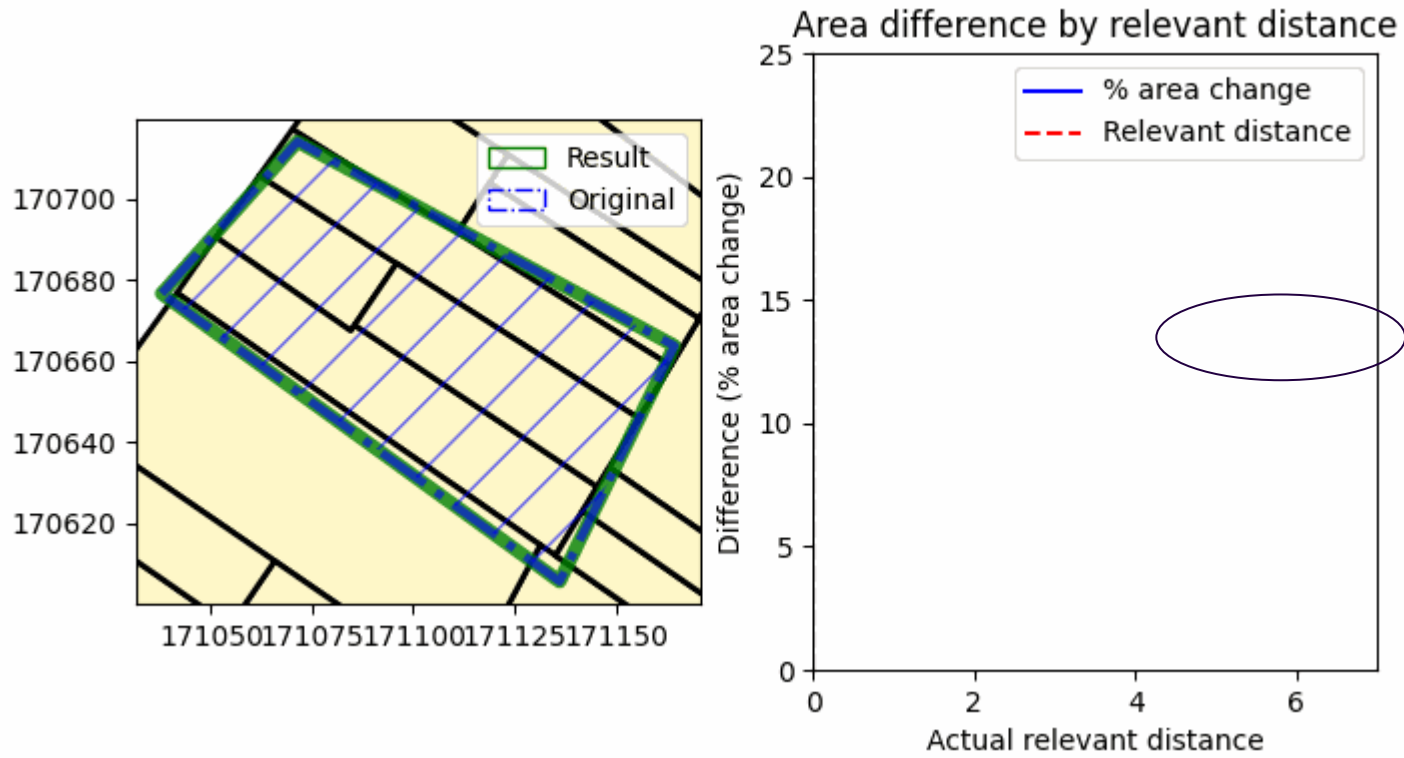
Relevant distance = data/business dependent

- Dataset history
- Geometry source
- Accuracy
- Object detail level
- ...

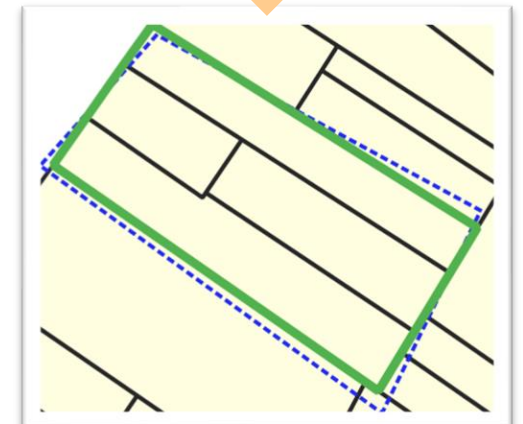
Alignment-algorithm (‘brdr’)

“How to determine this ‘relevant distance’?”

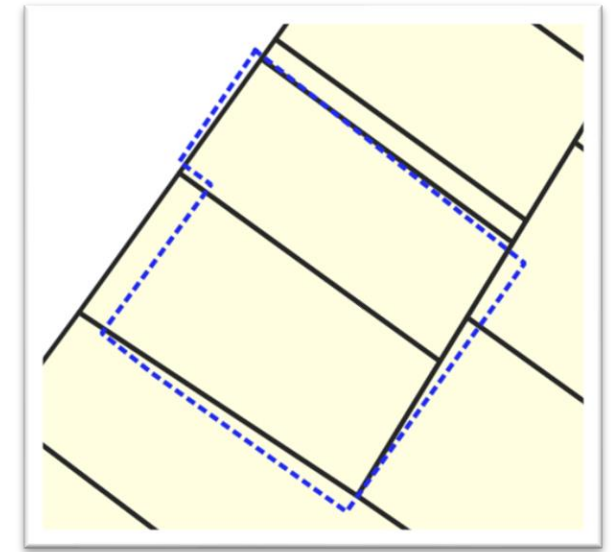
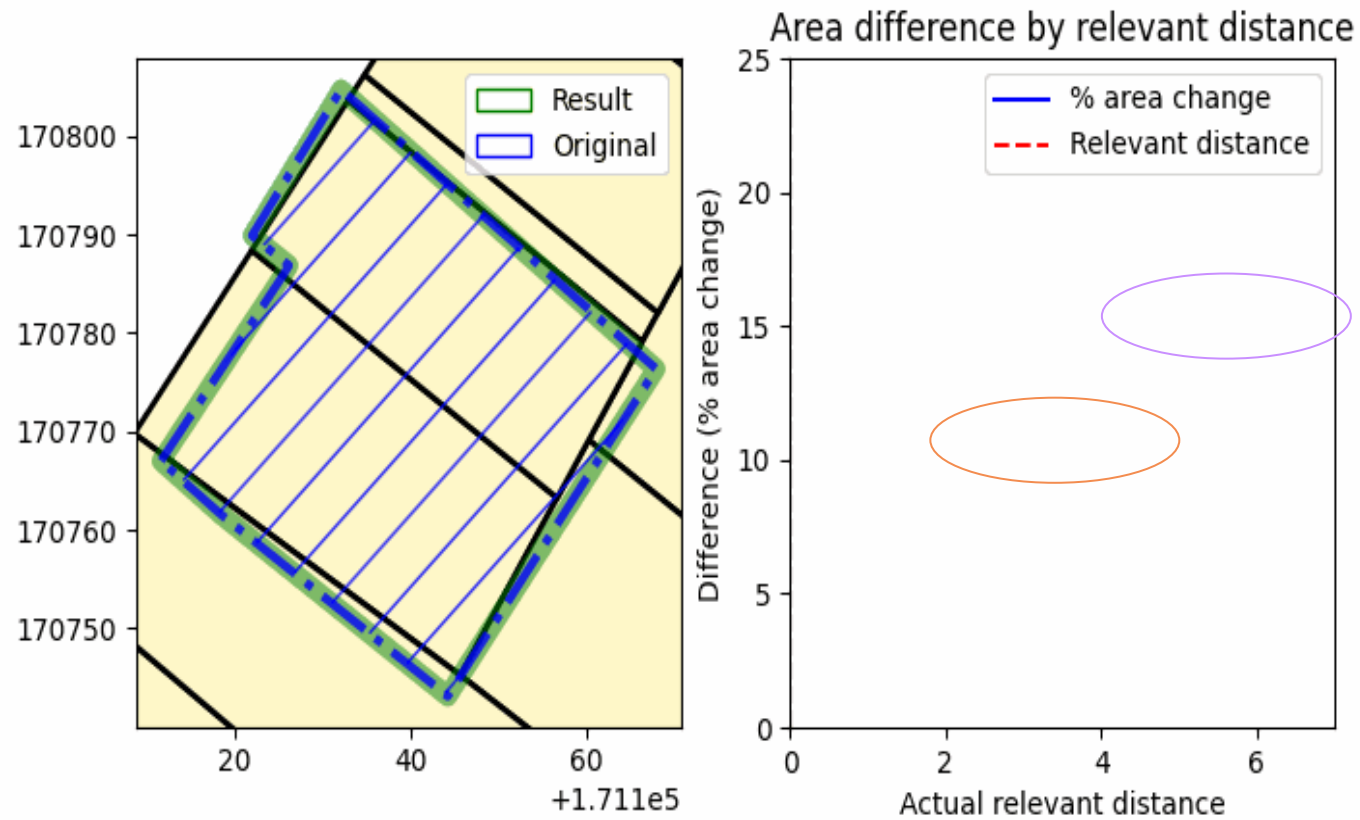
→ ‘brdr’ tries all !



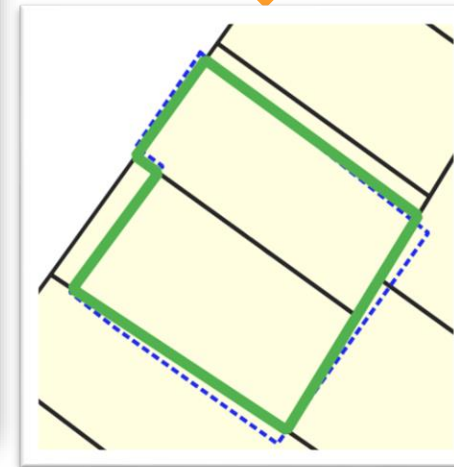
PREDICTION
@ 4.8m



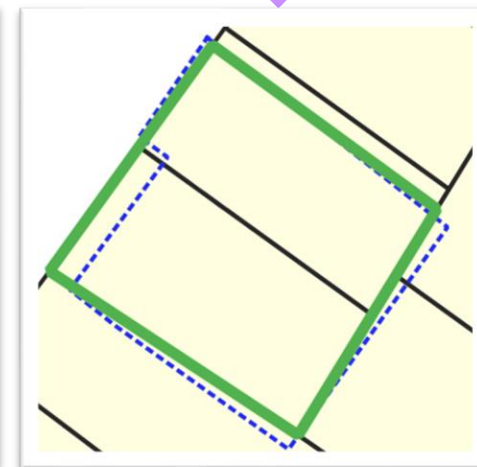
Alignment-algorithm (‘brdr’)



PREDICTION 1
@ 2.8m



PREDICTION 2
@ 4.4m



Predictions

Executing the algorithm with a series of 'relevant distances':

- 0,0.1,0.2,...1...1.9,2

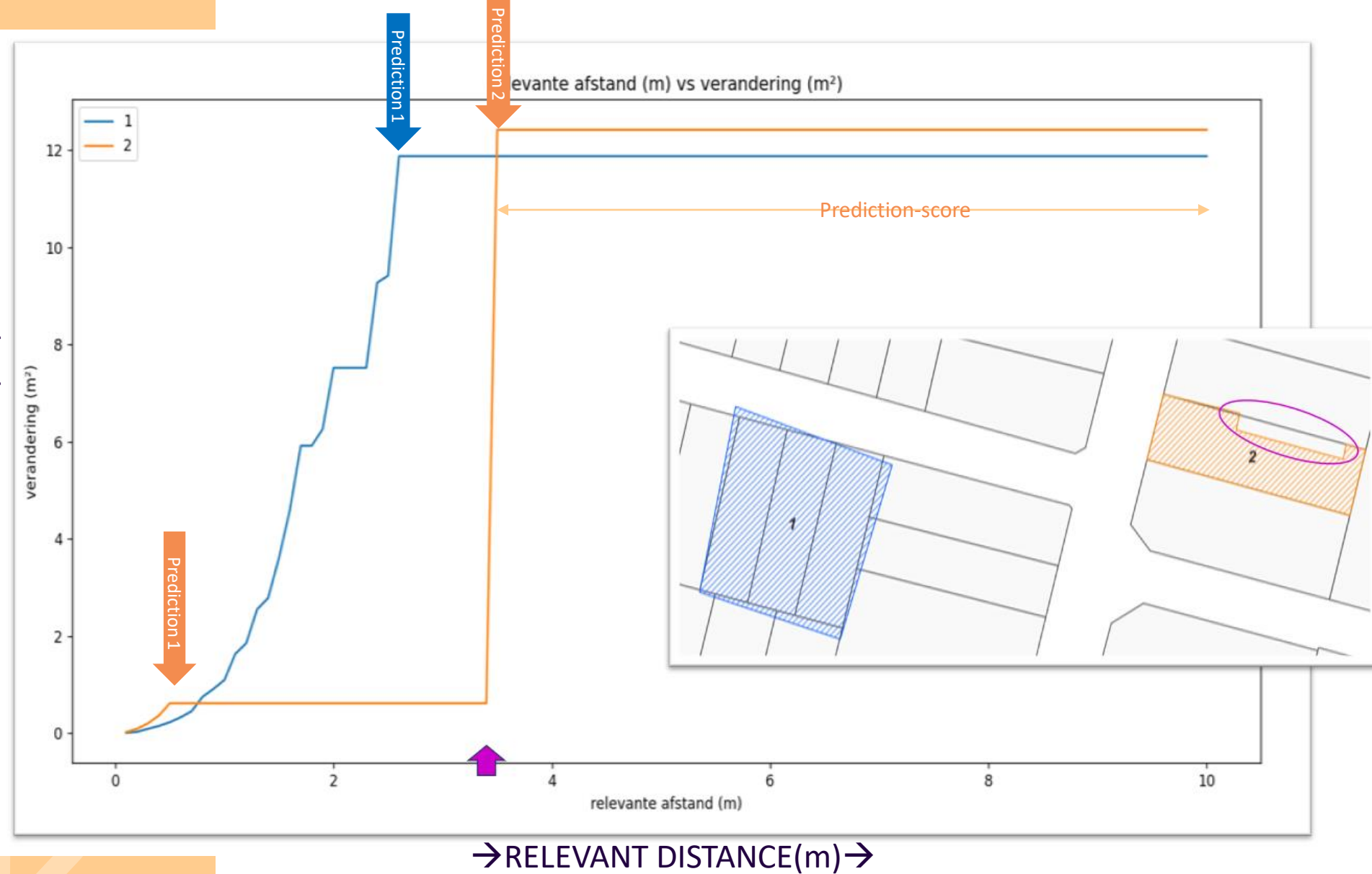
Change measurement (m^2) =
Difference in area between input and resulting geometry

PREDICTIONS=
'Stable' results

PREDICTION SCORE=

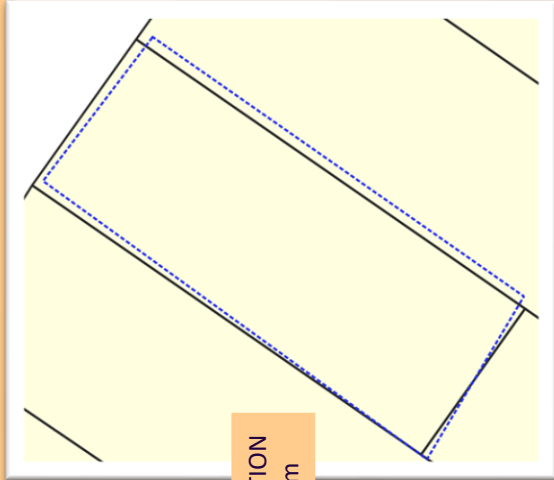
Length of stability

→ CHANGE (m^2) →

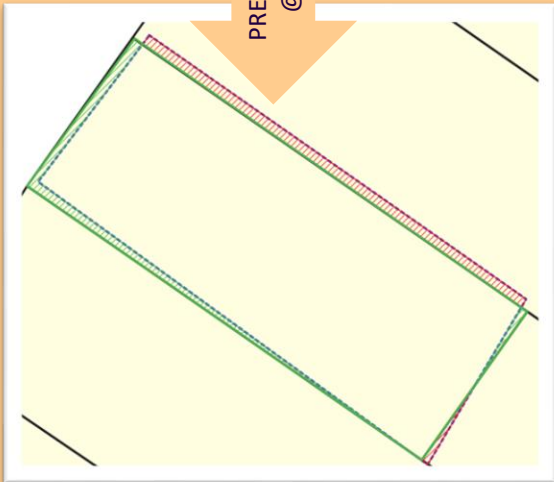


Basic Example

(Code 'brdr' v0.14.0)



PREDICTION
@1.2m



```
# CREATE AN ALIGNER
aligner = Aligner(crs="EPSG:31370")

# Load thematic WKT into Aligner
thematic_data = {
    "theme_id_1": geom_from_wkt("POLYGON ((170996.19647056088433601 170611.472792530228616
})
thematic_loader = DictLoader(thematic_data)
aligner.load_thematic_data(thematic_loader)

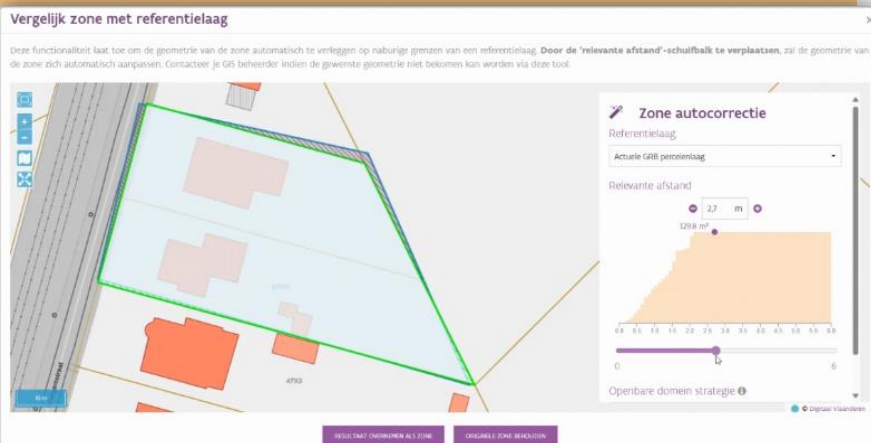
# Load reference data (GRB parcels)
reference_loader = GRBActualLoader(grb_type=GRBType.ADP, aligner=aligner)
aligner.load_reference_data(reference_loader)

# EXECUTE THE PREDICTOR
aligner.predictor()

#Get GeoJson-object of resulting prediction
featureclasses = aligner.get_results_as_geojson(resulttype=AlignerResultType.PREDICTIONS)
print (featureclasses['result'])
# SAVE PREDICTIONS TO file (GeoJSON)
aligner.save_results(path = 'output/', resulttype=AlignerResultType.PREDICTIONS)
```

'brdr' for geodata-management

Use cases



Implementation in business-processes, workflows and tooling (f.e brdrQ)

- Bulk geodata-alignment
- Feature-by-feature alignment: Predictions
- Geodata-lifecycle-support: Automatic actualizations to newer version of reference-data

Alignment after reprojection :

- f.e. Lambert72 -> Lambert2008

Data cleaning:

- sliver-cleanup and validity

Version management:

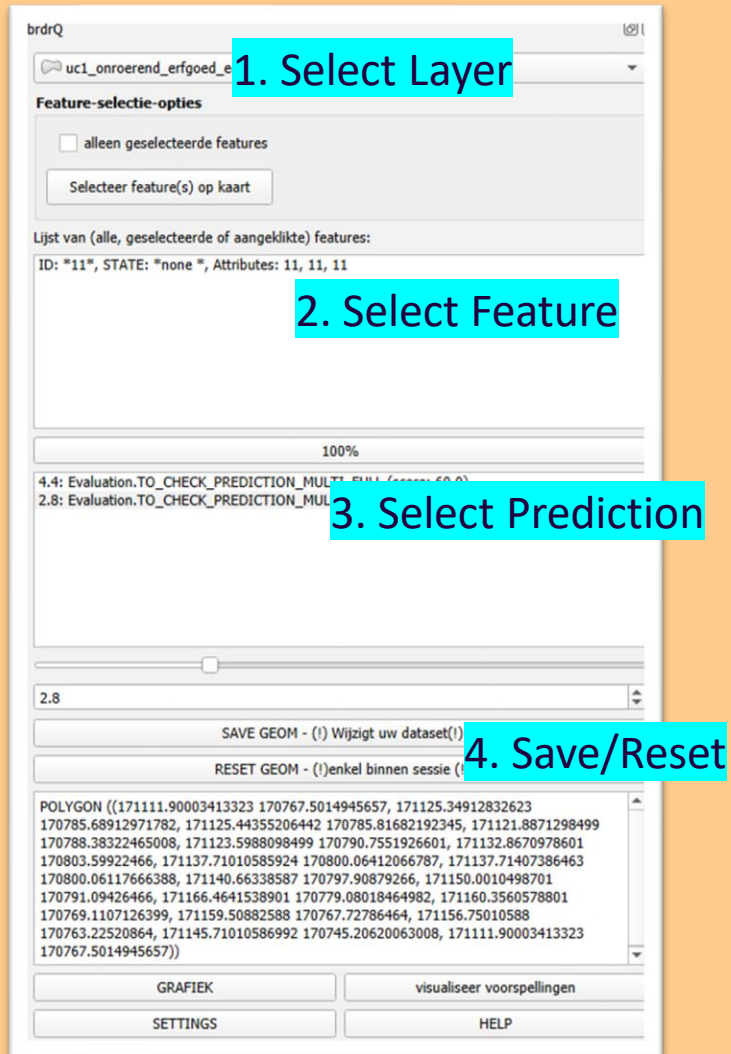
- Visualize differences between versions of geodata

Research & Data-Analysis:

- Investigate the pattern in deviation and change between thematic and reference boundaries

...

What is brdrQ?



✓ Plugin for QGIS

✓ UI to assist in aligning features

✓ Uses 'brdr' for processing

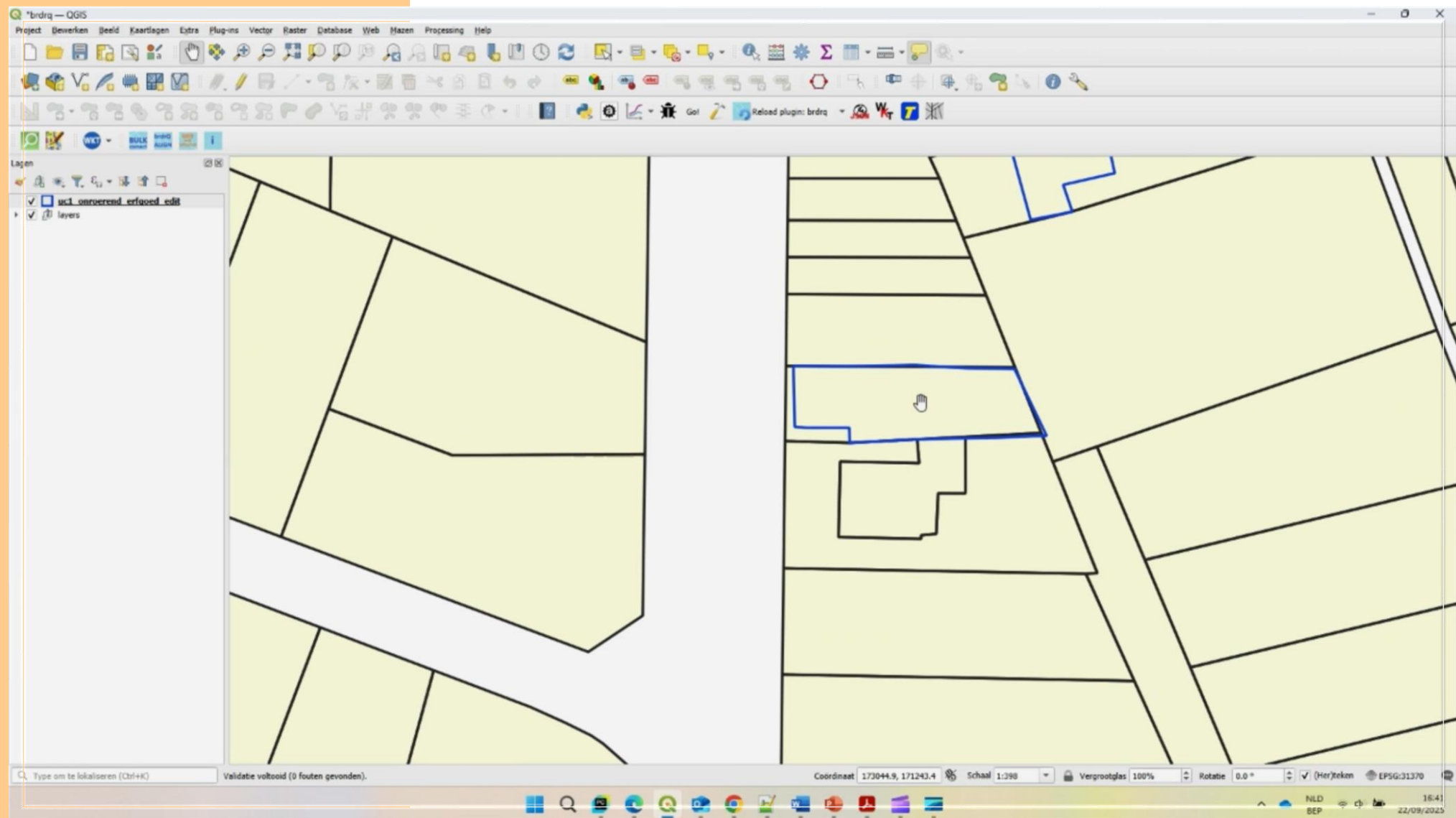
✓ Tools:

- Bulk alignment
- Feature-to-feature alignment (predictions)
- Settings: preferences, strategies,...

✓ Installation:

- see instructions on <https://github.com/OnroerendErfgoed/brdrQ>

brdrQ



Future



↳ brdr

- Optimization of 'brdr' code base
- Improvements based on user feedback
- Documentation
- Towards **v1.0.0**
- Enhancement: support for non-polygons

↳ brdrQ

- Improvements based on user feedback
- Documentation
- Towards **v1.0.0**

Thanks!

Questions?

Want to try?

- Python package 'brdr': <https://github.com/OnroerendErfgoed/brdr>
- QGIS – plugin brdrQ: <https://github.com/OnroerendErfgoed/brdrQ>
- Project page (dutch): <https://docs.athumi.eu/vip/help/grenzen-verleggen>
- Project report (dutch): <https://doi.org/10.55465/SXCW6218>

Bugs?

- Brdr – bug? → <https://github.com/OnroerendErfgoed/brdr/issues>
- BrdrQ – bug? → <https://github.com/OnroerendErfgoed/brdrQ/issues>
- Idea? → discussion forum: <https://github.com/OnroerendErfgoed/brdr/discussions/>

Ideas/Improvement/Discussion

- discussion forum: <https://github.com/OnroerendErfgoed/brdr/discussions/>
- E-mail: <karel.dieussaert@vlaanderen.be> <Karel.dieussaert@geosolutions.be>